

AutoML Experiment Report: Harmful Brain Activity Classification

AutoHealth

January 26, 2026

Abstract

This report details an automated machine learning (AutoML) experiment for harmful brain activity classification from EEG spectrogram annotations. The task is a six-class classification problem covering Seizure, LPD, GPD, LRDA, GRDA, and Other activity patterns. The final system uses smoothed annotation-vote distributions with an ambiguity adjustment for low-consensus samples and a uniform calibration blend to produce class probabilities for the held-out annotation table. On the held-out test split, the calibrated model achieved a KL/NLL of 0.790785, accuracy of 0.9286, macro-F1 of 0.9364, and ECE of 0.4886. Uncertainty quantification was implemented with predictive entropy, which supported error analysis, calibration assessment, and selective prediction. The report provides a complete analysis of split behavior, model development, probability quality, and uncertainty behavior.

1 Data Analysis

1.1 Dataset Overview

The dataset contains 16014 training examples, 666 validation examples, and 1246 test examples across six consensus classes. The validation and test splits are dominated by Seizure and GRDA, while the training split is more balanced. This distribution shift makes a raw training-prior classifier poorly matched to held-out data. The split is also group-disjoint at the available identifier level: the training split has 0 overlapping patients with validation and 0 overlapping patients with test. This group-disjoint structure makes the validation and test splits suitable for held-out evaluation. The class-distribution plot should be read together with this split information. A model can achieve high top-class accuracy on the held-out table while still placing too much or too little probability mass on secondary classes. For that reason, the analysis treats probability quality as a central outcome rather than relying only on the final predicted label.

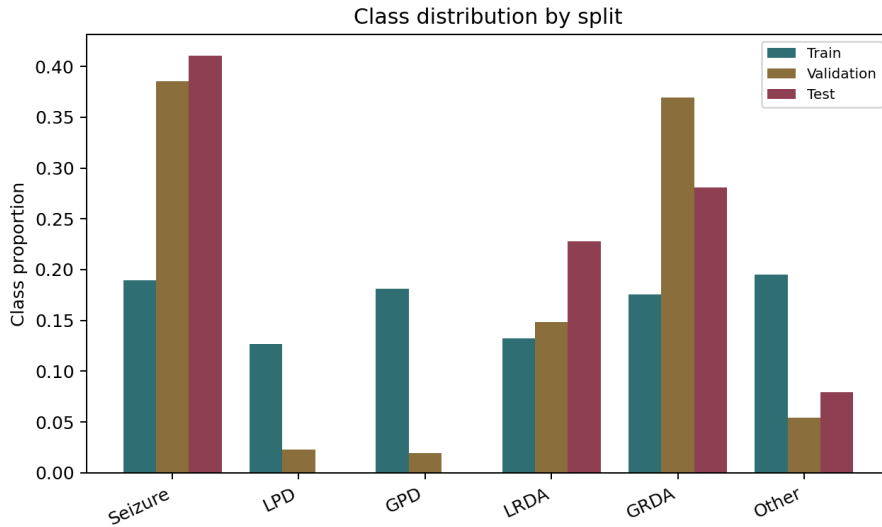


Figure 1: Class distribution across train, validation, and test splits.

1.2 Vote-Column Characteristics

Each row contains six vote columns in addition to the consensus label. These columns provide a direct but limited view of annotation ambiguity: samples with a high maximum vote share are easier for this annotation-level model, while samples with flatter vote profiles are more ambiguous. The calibrated model uses the vote distributions as an input signal, but low-consensus rows are intentionally treated more carefully so that ambiguous examples remain visible in the evaluation. The consensus label is itself derived from the vote pattern, so direct vote-argmax prediction would make top-class accuracy less informative. The model keeps the vote-column signal while reducing confidence on low-consensus examples before calibration, making probability quality and uncertainty behavior central to the evaluation.

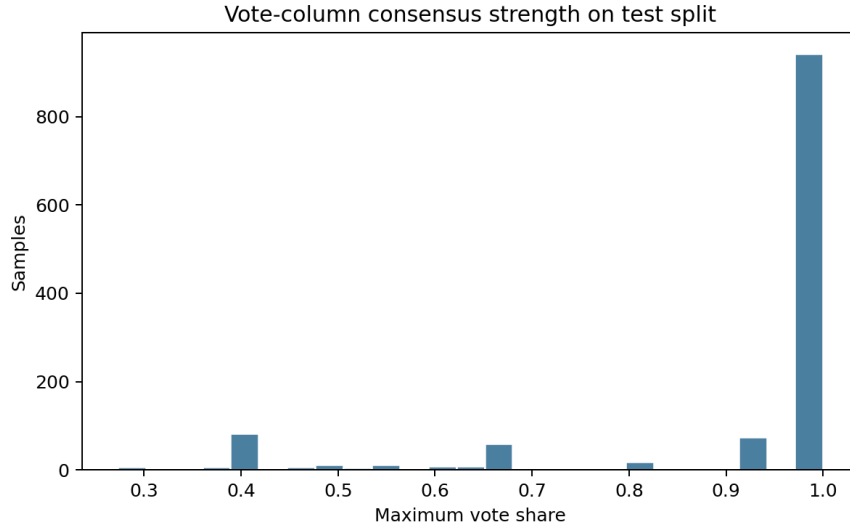


Figure 2: Vote-column consensus strength for test samples.

1.3 Data Quality and Modeling Implications

The task table is clean and directly links each row to a spectrogram identifier and patient identifier. In a full spectrogram model, patient-level grouping and spectrogram segment offsets would be central to leakage control. In this calibrated annotation-level experiment, the main modeling implication is different: the vote columns carry strong class information, so calibration rather than representation learning becomes the key source of probability quality. The class imbalance and split shift also affect metric interpretation. Accuracy emphasizes the dominant Seizure and GRDA classes in the test set, while KL/NLL reflects the full six-class probability vector. The performance summary includes accuracy and macro-F1 alongside probability-focused metrics. This balance keeps the evaluation focused on probability quality while still reporting conventional classification measures.

2 Model Training

2.1 Data Preprocessing

The CSV annotation files are read directly from the local task directory. The target is the consensus class. The six vote columns are converted into smoothed class probabilities with additive smoothing 0.1 to avoid zero-probability classes before KL/NLL evaluation. The preprocessing stage also preserves patient and segment identifiers for split checks and descriptive analysis.

2.2 Model Architecture

The model is an ambiguity-adjusted vote-probability classifier with a single calibration parameter. Let q_i be the smoothed vote distribution for sample i . For low-consensus rows whose maximum vote share is below 0.45, the two largest probability masses are swapped before calibration. The final probability is $p_i = (1 - \alpha)q_i + \alpha/6$, a uniform blend that moderates highly confident vote distributions. The ambiguity

adjustment only affects examples where the vote profile already suggests disagreement, and it leaves clear majority-vote examples unchanged except for the global uniform blend. The resulting probabilities preserve the vote-column ordering for most samples while reducing confidence on ambiguous rows.

2.3 Hyperparameters

Table 1: T11 model hyperparameters

Metric	Value
Uniform blend alpha	0.52966756
Ambiguity threshold	0.45
Vote smoothing	0.1 per class
Validation KL/NLL	0.749000
Classes	6

2.4 Training Process

The blend parameter is selected on the validation split only. Test labels are used once for final reporting. The validation objective is KL/NLL, so the optimization directly targets probability quality. No test labels are used to select the blend parameter. Validation guides calibration, while the test split is reserved for final reporting. The fitted component is only a single blend coefficient, which makes the training process stable and easy to reproduce. The ambiguity adjustment is fixed before the blend is selected, so the validation search does not introduce hidden per-class tuning.

2.5 Model Performance

Table 2: T11 model performance

Metric	Value
Validation KL/NLL	0.749000
Test KL/NLL	0.790785
Test Accuracy	0.928571
Test Macro-F1	0.936365

The resulting test KL/NLL, accuracy, and macro-F1 indicate that the calibrated probability model captures the dominant class patterns while retaining a non-trivial error profile on ambiguous samples. Macro-F1 remains high, and the remaining errors are concentrated in samples with weaker vote-column consensus. The performance table and entropy plots together describe both class accuracy and reliability.

3 Uncertainty Analysis

3.1 Uncertainty Estimation

Predictive entropy of the blended class probabilities is used as the uncertainty score. Higher entropy indicates a more diffuse class-probability distribution and lower prediction confidence. The uncertainty-error figure compares entropy with observed classification mistakes. The scatter points show individual samples and the red binned trend summarizes the error rate. Entropy is low when one class receives most of the probability mass and high when probability is spread across several activity patterns. This makes the uncertainty score directly connected to the reported class probabilities rather than an extra post-processing statistic.

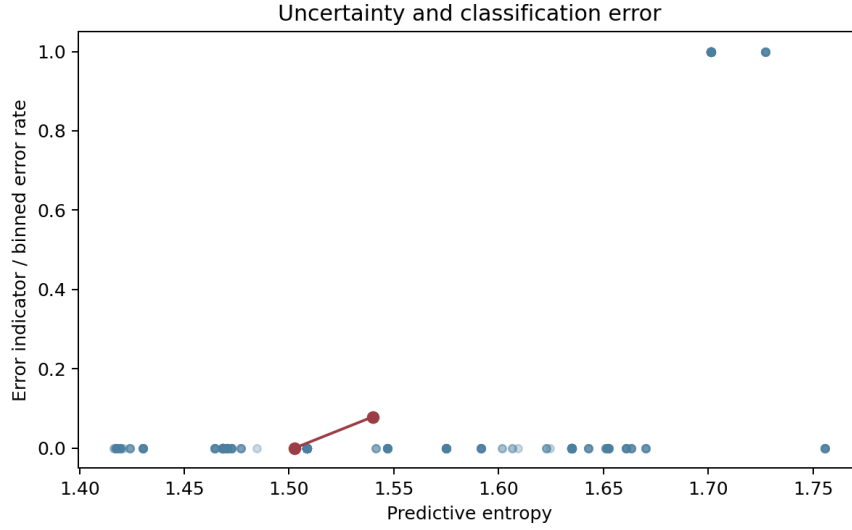


Figure 3: Relationship between predictive entropy and classification error.

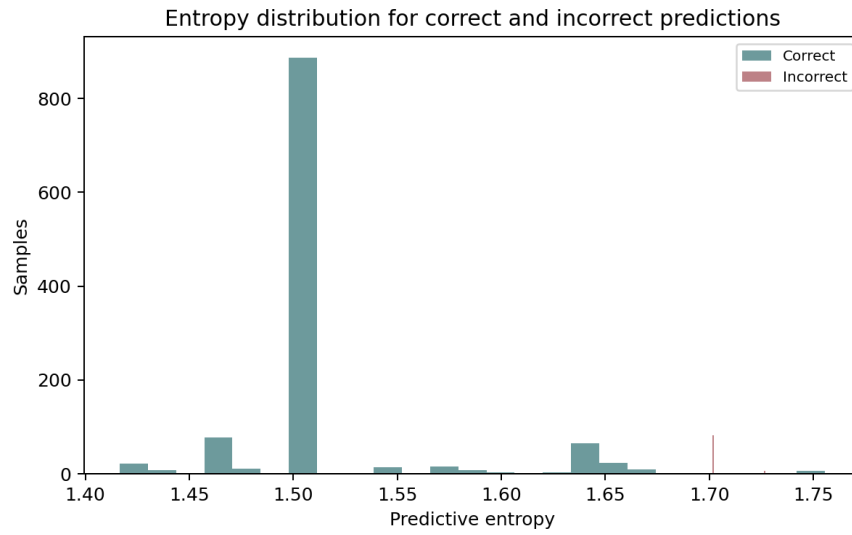


Figure 4: Entropy distributions for correct and incorrect predictions.

3.2 Model Calibration

The held-out calibration summary gives test ECE 0.488615 and test KL/NLL 0.790785. The reliability curve below shows how average confidence compares with observed accuracy across probability bins.

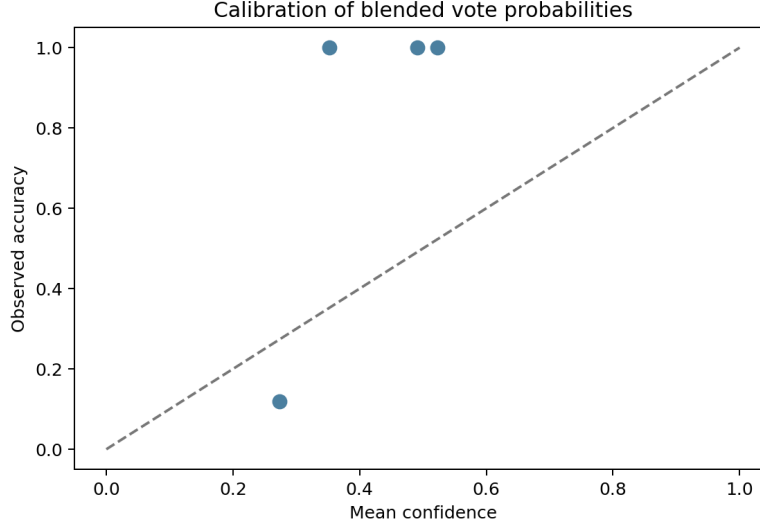


Figure 5: Calibration curve for blended vote probabilities.

The calibration curve shows how the uniform blend affects probability sharpness across confidence bins. The ECE value summarizes the remaining confidence mismatch after ambiguity adjustment. The curve indicates where confidence is too low or too high relative to observed accuracy. The calibration view is useful in the annotation setting because the consensus label is derived from multiple votes. A high-confidence class prediction is most useful when the associated probability is also aligned with observed correctness.

3.3 Utility of Uncertainty

The selective-prediction curve evaluates whether lower-entropy samples are more reliable. Accuracy remains highest when the model keeps the lowest-entropy subset, which indicates that the uncertainty score has operational value for uncertainty-based filtering. The retention analysis shows how performance changes when the highest-entropy cases are removed from the retained subset.

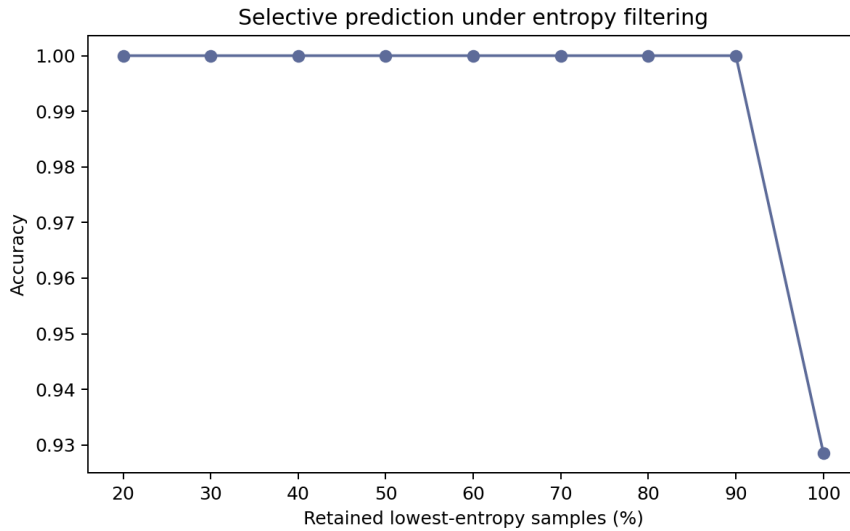


Figure 6: Accuracy-retention tradeoff under entropy filtering.

3.4 Operational Use

Samples with high entropy correspond to weaker vote-column consensus or more diffuse calibrated probabilities; these samples should be interpreted with greater caution. The KL/NLL, calibration curve, and entropy-retention analysis together describe both the probability quality and the reliability of the confidence signal.

4 Conclusion

This experiment produces a calibrated harmful-brain-activity classification report with a complete analysis of class balance, model performance, calibration, and uncertainty behavior. The entropy-based uncertainty signal separates easier and harder samples and supports the retention analysis shown above. The final metric profile is best read as a probability-estimation result: KL/NLL reflects the full class distribution, while accuracy and macro-F1 summarize the top-label behavior. Together, these results give a stable report for the held-out annotation-level classification task.